

Bounded solutions for general time interval BSDEs with quadratic growth coefficients and stochastic conditions

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This paper deals with bounded solutions for general time interval one-dimensional backward stochastic differential equations (BSDEs for short) with quadratic growth coefficients and stochastic conditions. Several general results of existence, uniqueness, stability and comparison for the bounded solutions are put forward and established, which improve considerably some existing works, even though for the case of finite time interval. Some new ideas are also developed to establish these results.

Keywords: Backward stochastic differential equation; general time interval; bounded solution; stochastic condition; quadratic growth.

AMS Subject Classification: 60H10

1. Introduction

Throughout this paper, let us always fix a positive integer d and a positive real number T which can be finite or infinite. Let $(\Omega, \mathcal{F}, P)$ be a complete probability space carrying a standard d -dimensional Brownian motion $(B_t)_{t \geq 0}$, $(\mathcal{F}_t)_{t \geq 0}$ the completed natural σ -algebra generated by $(B_t)_{t \geq 0}$ and $\mathcal{F}_T := \mathcal{F}$. We consider the following one-dimensional backward stochastic differential equation (BSDE):

$$y_t = \xi + \int_t^T g(s, y_s, z_s) ds - \int_t^T z_s \cdot dB_s, \quad t \in [0, T], \quad (1.1)$$

where T is called the time terminal, ξ is a $\mathcal{F}_T$ -measurable random variable called the terminal condition, and the random function $g(\omega, t, y, z) : \Omega \times [0, T] \times \mathbf{R} \times \mathbf{R}^d \mapsto \mathbf{R}$ is $(\mathcal{F}_t)$ -progressively measurable for each (y, z) and almost everywhere

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continuous with respect to (y, z) , called the generator. The triple (ξ, T, g) is called the parameters of BSDE (1.1), and usually denoted by BSDE (ξ, g) the BSDE whose generator is g and whose terminal condition is ξ . The solution of BSDE (1.1) is a pair of $(\mathcal{F}_t)$ -progressively measurable processes $(y_t, z_t)_{t \in [0, T]}$.

The nonlinear version of multidimensional BSDEs was first introduced by Pardoux and Peng [19], which established an existence and uniqueness result of square-integral solutions for BSDE (1.1) under the uniform Lipschitz condition of the generator g . Since then, BSDEs have been extensively studied due to its applications in many fields such as stochastic games, optimal control, mathematical finance and PDEs and so on, see [2, 11, 18, 20]. And many efforts have been made to improve the result of Pardoux and Peng [19] by weakening the uniform Lipschitz condition of the generator g , relaxing the finite time terminal T to a stopping time or infinity and generalizing the terminal condition ξ to non-square integrable case, see [1–10, 12, 13, 16, 17, 21] etc.

In particular, we would like to mention the following works related closely to our works. First of all, in 1997 El Karoui and Huang [7] introduced, at the first time, a kind of stochastic Lipschitz condition of the generator g in (y, z) , and established a general existence and uniqueness result for solutions of BSDE (1.1). This kind of condition was also employed in Bender and Kohlmann [1] and Wang *et al.* [21]. After that, in 2008 Briand and Confortola [2] put forward another kind of stochastic Lipschitz condition involving a BMO-martingale and obtained an existence and uniqueness result for solutions of BSDE (1.1). However, these works need some additional and stronger integrability assumptions with respect to the parameters (ξ, g) and the solutions. Recently, Liu *et al.* [17] eliminated these additional assumptions and established several existence, uniqueness and comparison results of L^p ($p > 1$) solutions for BSDEs (see Lemmas 2.1 and 2.2 for details), where the generator g satisfies some new stochastic conditions. On the other hand, we have to mention that Kobylanski [15] first introduced the quadratic growth condition of the generator g in z and investigated existence, uniqueness and stability of the bounded solutions for BSDE (1.1). And, BSDE with quadratic growth generator has attracted more and more interest in recent years due to its close connections with PDEs and mathematical finance, see, for example, Briand and Hu [4, 5], Delbaen *et al.* [6], Fan [9], Hu and Tang [12], Lepeltier and San Martin [16] and Morlais [18] etc.

Enlightened by these works, this paper is devoted to the bounded solutions of general time interval BSDEs with quadratic growth generators and some certain stochastic conditions. Several general results of existence, uniqueness, comparison and stability for the bounded solutions are established, which improve considerably some existing results, even though for the case of finite time interval. In addition, we also point out that under our weaker assumptions, some of the ideas and techniques employed in previous works are not valid. Consequently, we have to develop some new ideas to solve our problems.

The remaining of this paper is organized as follows. In Sec. 2, we introduce some basic notations, definitions and three lemmas. Section 3 aims at establishing a general monotonic stability result of bounded solutions for BSDEs (see Proposition 3.1), which is also a powerful tool to prove the existence of the solutions and generalizes the corresponding part of Theorem 2 in Lepeltier and San Martin [16] as well as Proposition 2.4 in Kobylanski [15] (see Remark 3.1). And, Sec. 4 is devoted to the uniqueness of the bounded solutions by establishing a general comparison theorem for BSDEs with quadratic growth generators and stochastic conditions (see Theorem 4.1), which improves considerably the corresponding comparison results obtained in Briand and Hu [5] and Fan [9] (see Remark 4.3). Based on Proposition 3.1 and Theorem 4.1 together with the techniques of truncation, convolution, exponential transform and weak convergence, in Sec. 5 we put forward and prove a general existence result of the bounded solutions for BSDEs (see Theorem 5.1), where the generator g has a stochastic semi-linear growth in y and a quadratic growth in z . This generalizes, to some extent, some corresponding existence results for the bounded solutions of BSDEs with quadratic growth coefficients obtained respectively in Lepeltier and San Martin [16], Kobylanski [15], Briand and Hu [4, 5] and Fan [9]. Here, we specially mention that the usual PDE-based or ODE-based ideas employed in Lepeltier and San Martin [16], Kobylanski [15] and Fan [9] are not effective anymore in our stochastic framework. Finally, in Sec. 6 we establish a general existence result of the minimal (resp. maximal) bounded solution for BSDEs with certain quadratic growth generators and stochastic conditions (see Theorem 6.1), which generalizes partly Theorem 3.1 in Fan [9] and Theorem 2 in Lepeltier and San Martin [16]. Several examples are also provided in Secs. 5 and 6 to illustrate that our results improve some existing works.

2. Notations, Definitions and Lemmas

Let us first introduce some notations which will be used in this paper. Denote $a \wedge b := \min\{a, b\}$, $a \vee b := \max\{a, b\}$, $a^- := -(a \wedge 0)$, $a^+ := (-a)^-$ and $\mathbf{R}_+ := [0, +\infty)$. For each predictable subset A of $\Omega \times [0, T]$, let $\mathbf{1}_A$ equal to 1 when $(\omega, t) \in A$, and 0 otherwise, and for every positive integer n , let $|\cdot|$ represent the norm of Euclidean space $\mathbf{R}^n$. For each $p > 0$, by $L^p(\Omega, \mathcal{F}_T, P)$ we denote the set of $\mathcal{F}_T$ -measurable random variables ξ such that $\mathbb{E}[|\xi|^p] < +\infty$, and by $L^\infty(\Omega, \mathcal{F}_T, P)$ the set of $\mathcal{F}_T$ -measurable bounded random variables endowed with the supremum norm $\|X\|_\infty := \sup\{x : P(|X| > x) > 0\}$. Moreover, for each $p > 0$, denote by $\mathcal{S}^p$ the set of real-valued, $(\mathcal{F}_t)$ -adapted and continuous processes $(Y_t)_{t \in [0, T]}$ such that

$$\|Y\|_{\mathcal{S}^p} := \left(\mathbb{E} \left[\sup_{t \in [0, T]} |Y_t|^p \right] \right)^{1 \wedge \frac{1}{p}} < +\infty.$$

If $p > 1$, $\|\cdot\|_{\mathcal{S}^p}$ is a norm on $\mathcal{S}^p$ and if $p \in (0, 1)$, $(X, X') \mapsto \|X - X'\|_{\mathcal{S}^p}$ defines a distance on $\mathcal{S}^p$. Under this metric, $\mathcal{S}^p$ is complete. Denote by $\mathcal{S}^\infty$ the set of

$(\mathcal{F}_t)$ -adapted, continuous and bounded processes. We also use M^2 to denote the set of $(\mathcal{F}_t)$ -progressively measurable $\mathbf{R}^d$ -valued processes $(Z_t)_{t \in [0, T]}$ such that

$$\|Z\|_{M^2} := \left(\mathbb{E} \left[\int_0^T |Z_t|^2 dt \right] \right)^{\frac{1}{2}} < +\infty.$$

It is clear that M^2 is a Hilbert space endowed with this norm.

Let us recall that a local martingale $\int_0^\cdot z_s \cdot dB_s$ is a martingale of bounded mean oscillation, briefly called BMO-martingale, if and only if

$$\sup_{\tau \in \Sigma_T} \left\| \mathbb{E} \left[\int_\tau^T |z_s|^2 ds \middle| \mathcal{F}_\tau \right] \right\|_\infty < +\infty,$$

where Σ_T represents the set of all $(\mathcal{F}_t)$ -stopping time τ such that $0 \leq \tau \leq T$.

Finally, for each $p \geq 1$, let $L^p([0, T]; \mathbf{R}_+)$ represent a set of deterministic functions $u(\cdot) : [0, T] \mapsto \mathbf{R}_+$ satisfying

$$\int_0^T u^p(t) dt < +\infty,$$

and $L^\infty(\Omega; L^p([0, T]; \mathbf{R}_+))$ the set of $(\mathcal{F}_t)$ -progressively measurable processes $\bar{u}_t(\omega) : \Omega \times [0, T] \mapsto \mathbf{R}_+$ such that

$$\left\| \int_0^T \bar{u}_s^p(\omega) ds \right\|_\infty < +\infty.$$

In this paper, we use the following definitions with respect to the solution of BSDE (1.1).

Definition 2.1. A pair of $(\mathcal{F}_t)$ -progressively measurable processes $(y_t, z_t)_{t \in [0, T]}$ with values in $\mathbf{R} \times \mathbf{R}^d$ is called a solution of BSDE (1.1) if dP -a.s., $t \mapsto y_t$ is continuous, $t \mapsto z_t$ belongs to $L^2(0, T)$, $t \mapsto g(t, y_t, z_t)$ belongs to $L^1(0, T)$, and (1.1) holds true for each $t \in [0, T]$.

Definition 2.2. A pair of $(\mathcal{F}_t)$ -progressively measurable processes $(y_t, z_t)_{t \in [0, T]}$ is called a L^2 (resp. bounded) solution of BSDE (1.1) if it is a solution of BSDE (1.1) and belongs to $\mathcal{S}^2 \times M^2$ (resp. $\mathcal{S}^\infty \times M^2$).

Definition 2.3. We call $(y_t, z_t)_{t \in [0, T]}$ the minimal (resp. maximal) L^2 solution of BSDE (1.1) if it is a L^2 solution of BSDE (1.1) and $y \leq y'$ (resp. $y \geq y'$) for any L^2 solution (y', z') of BSDE (1.1). Similarly, we can define the minimal (resp. maximal) bounded solution of BSDE (1.1).

At the end of this section, we introduce three lemmas which will be used several times later. Firstly, the following Lemma 2.1 is a combination of Theorem 3.1 and Theorem 4.1 in Liu *et al.* [17].

Lemma 2.1. Assume that $0 < T \leq +\infty$, $\xi \in L^2(\Omega, \mathcal{F}_T, P)$ and the generator g satisfies the following two assumptions:

- (A1) There exist two stochastic processes $u, v \in L^\infty(\Omega; L^1([0, T]; \mathbf{R}_+))$ and $v \in L^\infty(\Omega; L^2([0, T]; \mathbf{R}_+))$ such that $dP \times dt$ -a.e., $\forall y_1, y_2 \in \mathbf{R}$ and $z_1, z_2 \in \mathbf{R}^d$,

$$|g(\omega, t, y_1, z_1) - g(\omega, t, y_2, z_2)| \leq u_t(\omega)|y_1 - y_2| + v_t(\omega)|z_1 - z_2|;$$

- (A2) $\mathbb{E}[(\int_0^T |g(\omega, t, 0, 0)| dt)^2] < +\infty$.

Then BSDE (ξ, g) admits a unique L^2 solution.

Furthermore, assume that $\xi, \xi' \in L^2(\Omega, \mathcal{F}_T, P)$, one of the generators g and g' satisfies (A1), and (y, z) and (y', z') are, respectively, L^2 solutions of BSDE (ξ, g) and BSDE (ξ', g') . If dP -a.s., $\xi \leq \xi'$ and $dP \times dt$ -a.e., $g(t, y, z) \leq g'(t, y, z)$ for each $(y, z) \in \mathbf{R}^{1+d}$, then for each $t \in [0, T]$, $y_t \leq y'_t$, dP -a.s.

In view of Theorem 5.1 in Liu *et al.* [17], in the same way as in Theorems 2.2 and 2.3 of Fan [9] we can prove the following existence and comparison result.

Lemma 2.2. Assume that $0 < T \leq +\infty$, $\xi \in L^2(\Omega, \mathcal{F}_T, P)$ and the generator g satisfies the following assumption:

- (A3) There exist three stochastic processes $u, f \in L^\infty(\Omega; L^1([0, T]; \mathbf{R}_+))$, $v \in L^\infty(\Omega; L^2([0, T]; \mathbf{R}_+))$ such that $dP \times dt$ -a.e., for each $(y, z) \in \mathbf{R}^{1+d}$,

$$|g(\omega, t, y, z)| \leq f_t(\omega) + u_t(\omega)|y| + v_t(\omega)|z|. \quad (2.1)$$

Then BSDE (ξ, g) admits both a minimal and a maximal L^2 solution.

Furthermore, assume that $\xi, \xi' \in L^2(\Omega, \mathcal{F}_T, P)$, g and g' are generators of BSDEs, and (y, z) and (y', z') are, respectively, L^2 solutions of BSDE (ξ, g) and BSDE (ξ', g') . If dP -a.s., $\xi \leq \xi'$, $dP \times dt$ -a.e., $g(t, y, z) \leq g'(t, y, z)$ for each $(y, z) \in \mathbf{R}^{1+d}$, and one of the following two statements holds true:

- (i) g satisfies (A3) and (y, z) is the minimal L^2 solution of BSDE (ξ, g) ;
- (ii) g' satisfies (A3) and (y', z') is the maximal L^2 solution of BSDE (ξ', g') ,

then for each $t \in [0, T]$, $y_t \leq y'_t$, dP -a.s.

Combining Lemma 3.4, Theorems 3.2 and 3.3 in Fan [9] yields the following lemma.

Lemma 2.3. Assume that $0 < T \leq +\infty$, $\xi \in L^\infty(\Omega, \mathcal{F}_T, P)$ and g is a generator satisfying

- (A4) There exist a constant $\gamma \in \mathbf{R}_+$ and a function $u(\cdot) \in L^1([0, T]; \mathbf{R}_+)$ such that $dP \times dt$ -a.e., for each $(y, z) \in \mathbf{R}^{1+d}$,

$$|g(\omega, t, y, z)| \leq u(t) + \frac{\gamma}{2}|z|^2.$$

Then BSDE (ξ, g) admits both a minimal and a maximal bounded solution.

Furthermore, assume that $\xi, \xi' \in L^\infty(\Omega, \mathcal{F}_T, P)$, both generators g and g' satisfy (A4), and (y, z) and (y', z') are, respectively, the minimal (resp. maximal) bounded solution of BSDE (ξ, g) and BSDE (ξ', g') . If dP -a.s., $\xi \leq \xi'$ and $dP \times dt$ -a.e., $g(t, y, z) \leq g'(t, y, z)$ for each $(y, z) \in \mathbf{R}^{1+d}$, then for each $t \in [0, T]$, $y_t \leq y'_t$, dP -a.s.

3. A Monotonic Stability Result of Bounded Solutions

In this section, by developing a method employed in Theorem 2 of Lepeltier and San Martin [16], we prove the following Proposition 3.1, which will play an important role in the proof of the existence result of bounded solutions.

Proposition 3.1. *Assume that $0 < T \leq +\infty$ and that for each $n \geq 1$, $\xi^n \in L^\infty(\Omega, \mathcal{F}_T, P)$, g_n is a generator and $(y_t^n, z_t^n)_{t \in [0, T]}$ is a bounded solution of the following BSDE:*

$$y_t^n = \xi^n + \int_t^T g_n(s, y_s^n, z_s^n) ds - \int_t^T z_s^n \cdot dB_s, \quad t \in [0, T].$$

Assume further that $dP \times dt$ -a.e., $y^n \uparrow y$ (resp. $y^n \downarrow y$) as $n \rightarrow +\infty$, and that there exist two constants $A, \mu \in \mathbf{R}_+$ and a $(\mathcal{F}_t)$ -progressively measurable nonnegative process $(f_t)_{t \in [0, T]}$ satisfying $\mathbb{E}[\int_0^T |f_t| dt] < +\infty$ such that $dP \times dt$ -a.e.,

$$\sup_{n \geq 1} |y_t^n(\omega)| \leq A \quad (3.1)$$

and

$$\forall (y, z) \in [-A, A] \times \mathbf{R}^d, \quad \sup_{n \geq 1} |g_n(\omega, t, y, z)| \leq f_t(\omega) + \mu |z|^2. \quad (3.2)$$

Then $\{(z_t^n)_{t \in [0, T]}\}_{n=1}^{+\infty}$ admits a convergence point z in M^2 . Furthermore, if the sequence of generators $\{g_n\}_{n=1}^{+\infty}$ converges locally uniformly to a generator g , i.e. for any sequence $\{(y^n, z^n)\}_{n=1}^{+\infty}$ in $\mathbf{R} \times \mathbf{R}^d$, $dP \times dt$ -a.e.,

$$g_n(t, y^n, z^n) \rightarrow g(t, y, z) \text{ provided that } (y^n, z^n) \rightarrow (y, z), \quad \text{as } n \rightarrow +\infty, \quad (3.3)$$

then $(y_t, z_t)_{t \in [0, T]}$ is a bounded solution of BSDE (ξ, g) with $\xi = y_T$.

To prove Proposition 3.1, we need the following Lemma 3.1.

Lemma 3.1. *Assume that $0 < T \leq +\infty$ and $\{(y_t^n)_{t \in [0, T]}\}_{n=1}^{+\infty}$ and $\{(z_t^n)_{t \in [0, T]}\}_{n=1}^{+\infty}$ are, respectively, a sequence of $(\mathcal{F}_t)$ -progressively measurable $\mathbf{R}$ -valued and $\mathbf{R}^d$ -valued stochastic processes such that $dP \times dt$ -a.e., $\sup_n |y_t^n| \leq K$ and $\sup_n \|z_t^n\|_{M^2} \leq K$ for some constant $K \in \mathbf{R}_+$. If $dP \times dt$ -a.e., $y^n \rightarrow y$ and $z^n \xrightarrow{w} z$ (in M^2) as $n \rightarrow +\infty$, then $y^n z^n \xrightarrow{w} y z$ (in M^2) as $n \rightarrow +\infty$.*

Proof. For any bounded linear functional φ in M^2 , by Riesz's theorem there exists a unique $f \in M^2$ such that $\varphi(h) = \mathbb{E}[\int_0^T \langle h_t, f_t \rangle dt]$ for each $h \in M^2$, where and hereafter $\langle x, x' \rangle$ represents the usual inner product between vectors x and x' valued

in $\mathbf{R}^d$. Thus,

$$\begin{aligned}
 & |\varphi(y^n z^n) - \varphi(y.z.)| \\
 &= |\varphi((y^n - y.)z^n) + \varphi(y.(z^n - z.))| \\
 &= \left| \mathbb{E} \left[\int_0^T \langle (y_t^n - y_t)z_t^n, f_t \rangle dt \right] + \mathbb{E} \left[\int_0^T \langle y_t(z_t^n - z_t), f_t \rangle dt \right] \right| \\
 &\leq \mathbb{E} \left[\int_0^T |y_t^n - y_t| |z_t^n| |f_t| dt \right] + \left| \mathbb{E} \left[\int_0^T \langle (z_t^n - z_t), y_t f_t \rangle dt \right] \right|. \quad (3.4)
 \end{aligned}$$

By Hölder's inequality,

$$\begin{aligned}
 \mathbb{E} \left[\int_0^T |y_t^n - y_t| |z_t^n| |f_t| dt \right] &\leq \left\{ \mathbb{E} \left[\int_0^T |y_t^n - y_t|^2 |f_t|^2 dt \right] \right\}^{\frac{1}{2}} \cdot \left\{ \mathbb{E} \left[\int_0^T |z_t^n|^2 dt \right] \right\}^{\frac{1}{2}} \\
 &\leq \left\{ \mathbb{E} \left[\int_0^T |y_t^n - y_t|^2 |f_t|^2 dt \right] \right\}^{\frac{1}{2}} \cdot \sup_{n \geq 1} \|z^n\|_{M^2}.
 \end{aligned}$$

In view of the facts that $dP \times dt$ -a.e., $\sup_{n \geq 1} |y^n| \leq K$, $y^n \rightarrow y$ as $n \rightarrow +\infty$, $f. \in M^2$ and $\sup_{n \geq 1} \|z^n\|_{M^2} \leq K$, Lebesgue's dominated convergence theorem yields that

$$\mathbb{E} \left[\int_0^T |y_t^n - y_t| |z_t^n| |f_t| dt \right] \rightarrow 0 \quad \text{as } n \rightarrow +\infty. \quad (3.5)$$

On the other hand, since $y.f. \in M^2$ and $z^n \xrightarrow{w} z$ (in M^2) as $n \rightarrow +\infty$, it follows that there exists a bounded linear functional $\tilde{\varphi} \in M^2$ such that

$$\mathbb{E} \left[\int_0^T \langle (z_t^n - z_t), y_t f_t \rangle dt \right] = \tilde{\varphi}(z^n - z.) \rightarrow 0 \quad \text{as } n \rightarrow +\infty. \quad (3.6)$$

Thus, in view of the definition of weak convergence, a combination of (3.4)–(3.6) yields that

$$y^n z^n \xrightarrow{w} y.z. \quad (\text{in } M^2) \quad \text{as } n \rightarrow +\infty.$$

The lemma is proved. $\square$

Now, we prove Proposition 3.1.

Proof. We only prove the case of $y^n \uparrow y.$, another case can be proved in the same way. The proof will be spilt into three steps.

First step. Let us prove that $\sup_{n \geq 1} \|z^n\|_{M^2} < +\infty$. Applying Itô's formula to $e^{3\mu y^n}$ and then taking expectation we have

$$\mathbb{E}[e^{3\mu y_0^n}] + \frac{1}{2} \mathbb{E} \left[\int_0^T 9\mu^2 e^{3\mu y_t^n} |z_t^n|^2 dt \right] = \mathbb{E}[e^{3\mu \xi^n}] + \mathbb{E} \left[\int_0^T 3\mu e^{3\mu y_t^n} g_n(t, y_t^n, z_t^n) dt \right].$$

Thus, it follows from (3.1) and (3.2) that for each $n \geq 1$,

$$\frac{3}{2}\mu^2\mathbb{E}\left[\int_0^T e^{3\mu y_t^n}|z_t^n|^2 dt\right] \leq \mathbb{E}[e^{3\mu A}] + 3\mu e^{3\mu A}\mathbb{E}\left[\int_0^T |f_t| dt\right],$$

from which with (3.1) we know the existence of a constant $C \in \mathbf{R}_+$ depending only on μ , A and f . such that

$$\sup_{n \geq 1} \|z^n\|_{M^2} \leq C. \quad (3.7)$$

Second step. Let us show that $\{z^n\}_n$ admits a convergence point z . in M^2 . First of all, take a weak convergence subsequence of $\{z^n\}_n$ which can still be denoted by itself, and assume that $z^n \xrightarrow{w} z$. (in M^2) as $n \rightarrow +\infty$. Then, applying Itô's formula to $\Psi(y^m - y^n)$ ($m > n$) with $\Psi(x) = \frac{1}{16\mu}(e^{16\mu x} - 1) - x$ and then taking expectation yield that

$$\begin{aligned} & \mathbb{E}[\Psi(y_0^m - y_0^n)] + \frac{1}{2}\mathbb{E}\left[\int_0^T \Psi''(y_t^m - y_t^n)|z_t^m - z_t^n|^2 dt\right] \\ &= \mathbb{E}\left[\int_0^T \Psi'(y_t^m - y_t^n)(g_m(t, y_t^m, z_t^m) - g_n(t, y_t^n, z_t^n)) dt\right]. \end{aligned}$$

Note that $\Psi(y^m - y^n) \geq \Psi(0) = 0$. In view of (3.1) and (3.2), by inequalities $(a+b+c)^2 \leq 3(a^2+b^2+c^2)$ and $(a+b)^2 \leq 2a^2+2b^2$, we deduce that $dP \times dt$ -a.e.,

$$\begin{aligned} |g_m(t, y_t^m, z_t^m) - g_n(t, y_t^n, z_t^n)| &\leq 2f_t + \mu|z_t^n|^2 + \mu|z_t^m|^2 \\ &\leq 2f_t + 3\mu|z_t^m - z_t^n|^2 + 5\mu|z_t - z_t^n|^2 + 5\mu|z_t|^2, \end{aligned}$$

and then

$$\begin{aligned} & \mathbb{E}\left[\int_0^T \left(\frac{\Psi''}{2} - 3\mu\Psi'\right)(y_t^m - y_t^n)|z_t^m - z_t^n|^2 dt\right] \\ &\leq \mathbb{E}\left[\int_0^T \Psi'(y_t^m - y_t^n)\{2f_t + 5\mu|z_t - z_t^n|^2 + 5\mu|z_t|^2\} dt\right]. \end{aligned} \quad (3.8)$$

In the sequel, noticing that $(\frac{\Psi''}{2} - 3\mu\Psi')(x) = 5\mu e^{16\mu x} + 3\mu$ and $dP \times dt$ -a.e., $y_t^m \uparrow y$. as $m \rightarrow +\infty$, we have that for each $n \geq 1$, $dP \times dt$ -a.e.,

$$\left[\left(\frac{\Psi''}{2} - 3\mu\Psi'\right)(y_t^m - y_t^n)\right]^{\frac{1}{2}} \rightarrow \left[\left(\frac{\Psi''}{2} - 3\mu\Psi'\right)(y_t - y_t^n)\right]^{\frac{1}{2}} \quad \text{as } m \rightarrow +\infty,$$

from which together with (3.1), (3.7) and the fact that $z^m \xrightarrow{w} z$. (in M^2) as $m \rightarrow +\infty$, by Lemma 3.1 we know that for each $n \geq 1$, as $m \rightarrow +\infty$,

$$\begin{aligned} & \left[\left(\frac{\Psi''}{2} - 3\mu\Psi'\right)(y_t^m - y_t^n)\right]^{\frac{1}{2}} (z_t^m - z_t^n) \\ & \xrightarrow{w} \left[\left(\frac{\Psi''}{2} - 3\mu\Psi'\right)(y_t - y_t^n)\right]^{\frac{1}{2}} (z_t - z_t^n) \text{ (in } M^2). \end{aligned}$$

Thus, combining back to (3.8) we obtain that

$$\begin{aligned} & \mathbb{E} \left[\int_0^T \left(\frac{\Psi''}{2} - 3\mu\Psi' \right) (y_t - y_t^n) |z_t - z_t^n|^2 dt \right] \\ & \leq \lim_{m \rightarrow +\infty} \mathbb{E} \left[\int_0^T \left(\frac{\Psi''}{2} - 3\mu\Psi' \right) (y_t^m - y_t^n) |z_t^m - z_t^n|^2 dt \right] \\ & \leq \mathbb{E} \left[\int_0^T \Psi'(y_t - y_t^n) \{2f_t + 5\mu|z_t - z_t^n|^2 + 5\mu|z_t|^2\} dt \right]. \end{aligned}$$

Hence,

$$\begin{aligned} & \mathbb{E} \left[\int_0^T \left(\frac{\Psi''}{2} - 8\mu\Psi' \right) (y_t - y_t^n) |z_t - z_t^n|^2 dt \right] \\ & \leq \mathbb{E} \left[\int_0^T \Psi'(y_t - y_t^n) \{2f_t + 5\mu|z_t|^2\} dt \right], \end{aligned}$$

and then, since $(\frac{\Psi''}{2} - 8\mu\Psi')(x) = 8\mu$, we have

$$\mathbb{E} \left[\int_0^T 8\mu|z_t - z_t^n|^2 dt \right] \leq \mathbb{E} \left[\int_0^T \Psi'(y_t - y_t^n) \{2f_t + 5\mu|z_t|^2\} dt \right].$$

Finally, in view of the facts of $y^n \uparrow y$, $\mathbb{E}[\int_0^T |f_t| dt] < +\infty$ and $z. \in M^2$, Lebesgue's dominated convergence theorem yields that

$$\lim_{n \rightarrow +\infty} \mathbb{E} \left[\int_0^T |z_t^n - z_t|^2 dt \right] = 0. \quad (3.9)$$

Third step. Assume now that (3.3) holds true, we prove that $(y., z.)$ is a bounded solution of BSDE (ξ, g) . In fact, in view of (3.9), we can find a convergence subsequence of $\{z^n\}_n$, denoted by itself again, such that

$$dP \times dt\text{-a.e.}, \quad z^n \rightarrow z. \quad \text{as } n \rightarrow +\infty, \quad \text{and} \quad \mathbb{E} \left[\int_0^T |z_t^n - z_t|^2 dt \right] \leq \frac{1}{2^n}. \quad (3.10)$$

Then,

$$\mathbb{E} \left[\int_0^T \sup_{n \geq 1} |z_t^n - z_t|^2 dt \right] \leq \mathbb{E} \left[\int_0^T \sum_{n=1}^{+\infty} |z_t^n - z_t|^2 dt \right] \leq \sum_{n=1}^{+\infty} \frac{1}{2^n} < +\infty,$$

from which it follows that

$$\mathbb{E} \left[\int_0^T \sup_{n \geq 1} |z_t^n|^2 dt \right] \leq 2\mathbb{E} \left[\int_0^T \sup_{n \geq 1} |z_t^n - z_t|^2 dt \right] + 2\mathbb{E} \left[\int_0^T |z_t|^2 dt \right] < +\infty.$$

Furthermore, in view of $y^n \uparrow y$, (3.10) and (3.3), we get that

$$dP \times dt\text{-a.e.}, \quad g_n(t, y^n, z^n) \rightarrow g(t, y, z.) \quad \text{as } n \rightarrow +\infty.$$

On the other hand, by (3.1) and (3.2) we know that $dP \times dt$ -a.e.,

$$|g_n(t, y_t^n, z_t^n) - g(t, y_t, z_t)| \leq 2f_t + \mu|z_t^n|^2 + \mu|z_t|^2 \leq 2f_t + \mu \sup_{n \geq 1} |z_t^n|^2 + \mu|z_t|^2.$$

Thus, Lebesgue's dominated convergence theorem yields that

$$\lim_{n \rightarrow +\infty} \mathbb{E} \left[\int_0^T |g_n(t, y_t^n, z_t^n) - g(t, y_t, z_t)| dt \right] = 0. \quad (3.11)$$

It then follows from (3.9), (3.11) and the fact of $y^n \uparrow y$ that

$$\lim_{n \rightarrow +\infty} \mathbb{E} \left[\sup_{t \in [0, T]} |y_t^n - y_t| \right] = 0.$$

Thus, $y \in \mathcal{S}^\infty$ and $(y_t, z_t)_{t \in [0, T]}$ is indeed a bounded solution of BSDE (ξ, g) . The proof is complete. $\square$

Remark 3.1. Proposition 3.1 generalizes the corresponding part of Theorem 2 in Lepeltier and San Martin [16] as well as Proposition 2.4 in Kobylanski [15], where the terminal time T is finite and the process f in (3.2) is bounded by a constant or a deterministic function in $L^1([0, T]; \mathbf{R}_+)$.

4. Comparison Theorem of Bounded Solutions

In this section, we aim at establishing a general comparison theorem of bounded solutions which will be used several times later. We will use the following assumptions on the generator g , where $0 < T \leq +\infty$.

(H1) There exists a stochastic process $u \in L^\infty(\Omega; L^1([0, T]; \mathbf{R}_+))$ such that $dP \times dt$ -a.e., $\forall y_1, y_2 \in \mathbf{R}$ and $z \in \mathbf{R}^d$,

$$\text{sgn}(y_1 - y_2)(g(\omega, t, y_1, z) - g(\omega, t, y_2, z)) \leq u_t(\omega)|y_1 - y_2|.$$

(H2) There exist a continuous function $h(\cdot) : \mathbf{R} \mapsto \mathbf{R}_+$, a stochastic process $v \in L^\infty(\Omega; L^2([0, T]; \mathbf{R}_+))$, and a nondecreasing continuous function $\phi(\cdot) : \mathbf{R}_+ \mapsto \mathbf{R}_+$ with $\phi(0) = 0$ such that $dP \times dt$ -a.e., $\forall y \in \mathbf{R}$ and $z_1, z_2 \in \mathbf{R}^d$,

$$|g(\omega, t, y, z_1) - g(\omega, t, y, z_2)| \leq h(y)(|z_1| + |z_2|)|z_1 - z_2| + v_t(\omega)\phi(|z_1 - z_2|).$$

Here and henceforth we always assume that for all $x \in \mathbf{R}_+$, $0 \leq \phi(x) \leq ax + b$ where $a, b \in \mathbf{R}_+$. Furthermore, we also assume that $v \in L^\infty(\Omega; L^1([0, T]; \mathbf{R}_+))$ in the case of $b \neq 0$.

Remark 4.1. Assumption (H1) is first put forward and used in Liu *et al.* [17]. It is strictly weaker than the corresponding assumptions employed in several previous references including Morlais [18] and Fan [9], where $u_t(\omega)$ is a constant or a deterministic function in $L^1([0, T]; \mathbf{R}_+)$. The reader can be referred to Example 3.1 in Liu *et al.* [17] for more details. Similarly, generally speaking, we cannot suppose that the stochastic process $v_t(\omega)$ in (H2) is a deterministic function in $L^2([0, T]; \mathbf{R}_+)$.

Remark 4.2. Clearly, assumption (H2) generalizes and unifies the following two assumptions (H2a) and (H2b), which are employed in Briand and Elie [3], Fan [9], Fan *et al.* [10], Jia [13], Liu *et al.* [17] and Morlais [18] etc.

(H2a) There exists a stochastic process $v. \in L^\infty(\Omega; L^2([0, T]; \mathbf{R}_+))$ such that $dP \times dt$ -a.e., $\forall y \in \mathbf{R}$ and $z_1, z_2 \in \mathbf{R}^d$,

$$|g(\omega, t, y, z_1) - g(\omega, t, y, z_2)| \leq (v_t(\omega) + |z_1| + |z_2|)|z_1 - z_2|.$$

(H2b) There exist a stochastic process $v. \in L^\infty(\Omega; L^2([0, T]; \mathbf{R}_+))$ and a nondecreasing continuous function $\phi(\cdot) : \mathbf{R}_+ \mapsto \mathbf{R}_+$ with $\phi(0) = 0$ such that $dP \times dt$ -a.e., $\forall y \in \mathbf{R}$ and $z_1, z_2 \in \mathbf{R}^d$,

$$|g(\omega, t, y, z_1) - g(\omega, t, y, z_2)| \leq v_t(\omega)\phi(|z_1 - z_2|).$$

Here we always assume that for all $x \in \mathbf{R}_+$, $0 \leq \phi(x) \leq ax + b$ where $a, b \in \mathbf{R}_+$. Furthermore, we also assume that $v. \in L^\infty(\Omega; L^1([0, T]; \mathbf{R}_+))$ in the case of $b \neq 0$.

For example, let us consider the following generator

$$g(\omega, t, y, z) := \bar{v}_t(\omega)e^y|z|^\alpha + (\bar{v}_t(\omega))^{2-\beta}y^5|z|^\beta,$$

where $0 < \alpha < 1 \leq \beta < 2$ and $\bar{v}. \in L^\infty(\Omega; L^1([0, T]; \mathbf{R}_+)) \cap L^\infty(\Omega; L^2([0, T]; \mathbf{R}_+))$. Clearly, this generator g satisfies neither (H2a) nor (H2b). However, it satisfies (H2). In fact, by virtue of inequalities

$$||z_1|^\alpha - |z_2|^\alpha| \leq |z_1 - z_2|^\alpha$$

and

$$\begin{aligned} \bar{v}_t^{2-\beta}||z_1|^\beta - |z_2|^\beta| &\leq \beta \bar{v}_t^{2-\beta}(|z_1| + |z_2|)^{\beta-1}|z_1 - z_2| \\ &\leq \beta[(2-\beta)\bar{v}_t + (\beta-1)(|z_1| + |z_2|)]|z_1 - z_2|, \end{aligned}$$

we can check that this g satisfies assumption (H2) with $h(y) = e^y + |y|^5$, $v. = \bar{v}$. and $\phi(x) = x + x^\alpha$.

The following comparison theorem for bounded solutions of BSDEs is the main result of this section.

Theorem 4.1. (Comparison Theorem) *Assume that $0 < T \leq +\infty$, $\xi, \xi' \in L^\infty(\Omega, \mathcal{F}_T, P)$, both g and g' are two generators of BSDEs, and $(y., z.)$ and (y', z') are, respectively, a bounded solution of BSDE (ξ, g) and BSDE (ξ', g') such that $\int_0^\cdot z_s \cdot dB_s$ and $\int_0^\cdot z'_s \cdot dB_s$ are BMO-martingales. If dP -a.s., $\xi \leq \xi'$ and one of the following two statements holds true:*

- (i) *g satisfies (H1)–(H2), and $\mathbf{1}_{y_t > y'_t}(g(t, y'_t, z'_t) - g'(t, y'_t, z'_t)) \leq 0$, $dP \times dt$ -a.e.;*
- (ii) *g' satisfies (H1)–(H2), and $\mathbf{1}_{y_t > y'_t}(g(t, y_t, z_t) - g'(t, y_t, z_t)) \leq 0$, $dP \times dt$ -a.e.,*

then for each $t \in [0, T]$, $y_t \leq y'_t$, dP -a.s.

Proof. We only prove case (i). Another case can be proved in the same way. Suppose that g satisfies (H1) and (H2) with $u, v, h(\cdot)$ and $\phi(\cdot)$, $dP \times dt$ -a.e., $\mathbf{1}_{y_t > y'_t}(g(t, y'_t, z'_t) - g'(t, y'_t, z'_t)) \leq 0$ and $|y_t| + |y'_t| \leq A$ with $A \in \mathbf{R}_+$.

Set $\hat{y} := y - y'$, $\hat{z} := z - z'$ and $\hat{\xi} := \xi - \xi'$. Applying Itô's formula to $\hat{Y} := \hat{y} \cdot e^{\int_0^s u_s ds}$ yields that

$$\begin{aligned} \hat{Y}_t &= \hat{\xi} e^{\int_0^T u_t dt} + \int_t^T [-u_s \hat{Y}_s + e^{\int_0^s u_r dr} (g(s, y_s, z_s) - g'(s, y'_s, z'_s))] ds \\ &\quad - \int_t^T e^{\int_0^s u_r dr} \hat{z}_s \cdot dB_s, \quad t \in [0, T]. \end{aligned}$$

In view of dP -a.s., $\hat{\xi}^+ = 0$, Tanaka's formula leads to the following inequality with $t \in [0, T]$,

$$\begin{aligned} \hat{Y}_t^+ &\leq \int_t^T \mathbf{1}_{\hat{y}_s > 0} [-u_s \hat{Y}_s + e^{\int_0^s u_r dr} (g(s, y_s, z_s) - g'(s, y'_s, z'_s))] ds \\ &\quad - \int_t^T \mathbf{1}_{\hat{y}_s > 0} e^{\int_0^s u_r dr} \hat{z}_s \cdot dB_s. \end{aligned} \quad (4.1)$$

Since $\mathbf{1}_{\hat{y}_s > 0}(g(s, y'_s, z'_s) - g'(s, y'_s, z'_s))$ is non-positive, we have

$$\begin{aligned} \mathbf{1}_{\hat{y}_s > 0}(g(s, y_s, z_s) - g'(s, y'_s, z'_s)) \\ &= \mathbf{1}_{\hat{y}_s > 0}(g(s, y_s, z_s) - g(s, y'_s, z'_s)) + \mathbf{1}_{\hat{y}_s > 0}(g(s, y'_s, z'_s) - g'(s, y'_s, z'_s)) \\ &\leq \mathbf{1}_{\hat{y}_s > 0}(g(s, y_s, z_s) - g(s, y'_s, z_s)) + \mathbf{1}_{\hat{y}_s > 0}(g(s, y'_s, z_s) - g(s, y'_s, z'_s)) \end{aligned}$$

and we deduce, using assumptions (H1) and (H2) for g , that

$$\begin{aligned} \mathbf{1}_{\hat{y}_s > 0}(g(s, y_s, z_s) - g'(s, y'_s, z'_s)) \\ \leq u_s \hat{y}_s^+ + \mathbf{1}_{\hat{y}_s > 0} h(y'_s) \{(|z_s| + |z'_s|)|z_s - z'_s| + v_s \phi(|z_s - z'_s|)\}. \end{aligned} \quad (4.2)$$

In addition, from the proof of Theorem 1 in Fan *et al.* [10] we know that, with $c = a + b$,

$$\forall x \in \mathbf{R}_+ \quad \text{and} \quad n \geq 1, \quad \phi(x) \leq (n + 2c)x + \mathbf{1}_{b \neq 0} \phi\left(\frac{2c}{n + 2c}\right). \quad (4.3)$$

Combining (4.1)–(4.3) yields that for each $n \geq 1$ and $t \in [0, T]$,

$$\begin{aligned} \hat{Y}_t^+ &\leq a_n + \int_t^T \mathbf{1}_{\hat{y}_s > 0} \bar{K} e^{\int_0^s u_r dr} [(|z_s| + |z'_s|) + v_s(n + 2c)] |\hat{z}_s| ds \\ &\quad - \int_t^T \mathbf{1}_{\hat{y}_s > 0} e^{\int_0^s u_r dr} \hat{z}_s \cdot dB_s \\ &= a_n - \int_t^T \mathbf{1}_{\hat{y}_s > 0} e^{\int_0^s u_r dr} \hat{z}_s \cdot \left[dB_s - \frac{\bar{K}[(|z_s| + |z'_s|) + v_s(n + 2c)] \hat{z}_s}{|\hat{z}_s|} \mathbf{1}_{|\hat{z}_s| \neq 0} ds \right], \end{aligned} \quad (4.4)$$

where $\bar{K} := \max_{|y| \leq A} h(y)$ and

$$a_n := \mathbf{1}_{b \neq 0} e^{\|\int_0^T u_t dt\|_\infty} \bar{K} \phi\left(\frac{2c}{n+2c}\right) \cdot \left\| \int_0^T v_t dt \right\|_\infty \rightarrow 0 \quad \text{by (H2)} \quad \text{as } n \rightarrow +\infty. \quad (4.5)$$

Furthermore, it follows from the definition of BMO-martingale that for each $n \geq 1$,

$$\begin{aligned} & \sup_{\tau \in \Sigma_T} \left\| \mathbb{E} \left[\int_\tau^T \bar{K}^2 [(|z_s| + |z'_s|) + v_s(n+2c)]^2 ds \middle| \mathcal{F}_\tau \right] \right\|_\infty \\ & \leq 4\bar{K}^2(n+2c)^2 \left\| \int_0^T v_s^2 ds \right\|_\infty + 4\bar{K}^2 \sup_{\tau \in \Sigma_T} \left\| \mathbb{E} \left[\int_\tau^T |z_s|^2 ds \middle| \mathcal{F}_\tau \right] \right\|_\infty \\ & \quad + 4\bar{K}^2 \sup_{\tau \in \Sigma_T} \left\| \mathbb{E} \left[\int_\tau^T |z'_s|^2 ds \middle| \mathcal{F}_\tau \right] \right\|_\infty < +\infty, \end{aligned}$$

which means that for each $n \geq 1$, the process

$$M_n(t) := \int_0^t \frac{\bar{K}[(|z_s| + |z'_s|) + v_s(n+2c)] \hat{z}_s}{|\hat{z}_s|} \mathbf{1}_{|\hat{z}_s| \neq 0} \cdot dB_s, \quad t \in [0, T]$$

is a BMO-martingale under P . Then it follows from Theorem 2.3 in Kazamaki [14] that the stochastic exponential $\mathcal{E}(M_n)$ of M_n is a uniformly integrable martingale, where the stochastic exponential is denoted by

$$\mathcal{E}(M_n)(t) := \exp \left(M_n(t) - \frac{1}{2} \langle M_n \rangle_t \right), \quad t \in [0, T]$$

and the quadratic variation is denoted by $\langle M \rangle$. Now, let us define by P_n the probability measure under $(\Omega, \mathcal{F}_T)$ given by $\frac{dP_n}{dP} := \mathcal{E}(M_n)(T)$. Then, noticing that

$$\bar{M}_t := \int_0^t \mathbf{1}_{\hat{y}_s > 0} e^{\int_0^s u_r dr} \hat{z}_s \cdot dB_s, \quad t \in [0, T]$$

is also a BMO-martingale under P , by Theorem 3.6 in Kazamaki [14] we know that the process

$$\int_0^t \mathbf{1}_{\hat{y}_s > 0} e^{\int_0^s u_r dr} \hat{z}_s \cdot \left[dB_s - \frac{\bar{K}[(|z_s| + |z'_s|) + v_s(n+2c)] \hat{z}_s}{|\hat{z}_s|} \mathbf{1}_{|\hat{z}_s| \neq 0} ds \right], \quad t \in [0, T],$$

the Girsanov transform of $\bar{M}$, is a BMO-martingale under P_n .

Now, taking conditional expectation with respect to $\mathcal{F}_t$ under P_n in (4.4) yields that for each $t \in [0, T]$ and $n \geq 1$, $\hat{Y}_t^+ \leq a_n$, dP -a.s. Finally, letting $n \rightarrow +\infty$ in this inequality and in view of (4.5), we can conclude that for each $t \in [0, T]$, $y_t \leq y'_t$, dP -a.s. Theorem 4.1 is then proved. $\square$

Remark 4.3. In view of Remarks 4.1 and 4.2, Theorem 4.1 improves considerably the corresponding comparison results obtained in Briand and Hu [5] and Fan [9].

5. Existence and Uniqueness of Bounded Solutions

By virtue of the comparison result and stability result established in the previous two sections, in this section we will first prove a general existence result of bounded solutions for BSDEs (see Theorem 5.1). Then, based on Theorems 5.1 and 4.1, we will also give a new existence and uniqueness result of the bounded solution for BSDEs (see Theorem 5.2). First, we introduce the following assumptions with respect to the generator g :

(H3) There exist two stochastic processes $\bar{u}, f \in L^\infty(\Omega; L^1([0, T]; \mathbf{R}_+))$ and a constant $k \in \mathbf{R}_+$ such that $dP \times dt$ -a.e., for each $(y, z) \in \mathbf{R}^{1+d}$,

$$\operatorname{sgn}(y)g(\omega, t, y, z) \leq f_t(\omega) + \bar{u}_t(\omega)|y| + k|z|^2.$$

(H4) There exist a continuous function $\bar{h}(\cdot) : \mathbf{R} \mapsto \mathbf{R}_+$ and a $\psi(\cdot) \in \mathbf{S}$ such that $dP \times dt$ -a.e., for each $(y, z) \in \mathbf{R}^{1+d}$,

$$|g(\omega, t, y, z)| \leq \psi_t(\omega, |y|) + \bar{h}(y)|z|^2,$$

here and hereafter $\mathbf{S}$ denotes a set of functions $\psi_t(\omega, x) : \Omega \times [0, T] \times \mathbf{R}_+ \mapsto \mathbf{R}_+$ satisfying the following two conditions:

- $dP \times dt$ -a.e., the function $x \mapsto \psi_t(\omega, x)$ is increasing.
- for each $x \in \mathbf{R}_+$, $\psi_t(\omega, x)$ is a $(\mathcal{F}_t)$ -progressively measurable process with

$$\int_0^T \psi_t(\omega, x) dt < +\infty, \quad dP\text{-a.s.}$$

Remark 5.1. Assume that g is a generator with $g(\cdot, 0, 0) \in L^\infty(\Omega; L^1([0, T]; \mathbf{R}_+))$ and it satisfies assumptions (H1) and (H2), then g must satisfy assumption (H3). In fact, in view of (H1) and (H2), we have $dP \times dt$ -a.e., for each $y \in \mathbf{R}$ and $z \in \mathbf{R}^d$,

$$\begin{aligned} \operatorname{sgn}(y)g(\omega, t, y, z) &= \operatorname{sgn}(y)(g(\omega, t, y, z) - g(\omega, t, 0, z)) \\ &\quad + \operatorname{sgn}(y)(g(\omega, t, 0, z) - g(\omega, t, 0, 0)) + \operatorname{sgn}(y)g(\omega, t, 0, 0) \\ &\leq u_t(\omega)|y| + h(0)[|z|^2 + v_t(\omega) \cdot (a|z| + b)] + |g(\omega, t, 0, 0)| \\ &\leq f_t(\omega) + \bar{u}_t(\omega)|y| + k|z|^2 \end{aligned} \tag{5.1}$$

with $f_t(\omega) := |g(\omega, t, 0, 0)| + bh(0)v_t(\omega) + \frac{1}{2}h(0)a^2v_t^2(\omega)$, $\bar{u}_t(\omega) := u_t(\omega)$ and $k := \frac{3}{2}h(0)$, from which (H3) follows immediately.

In addition, note that g is continuous in (y, z) . Assumption (H3) can imply that

$$dP \times dt\text{-a.e.}, \quad |g(\omega, t, 0, 0)| \leq f_t(\omega),$$

and then $g(\cdot, 0, 0) \in L^\infty(\Omega; L^1([0, T]; \mathbf{R}_+))$.

Remark 5.2. It is completely possible that the function $\psi_t(\omega, x) \in \mathbf{S}$ defined in (H4) does not satisfy the following condition:

$$\mathbb{E} \left[\int_0^T \psi_t(\omega, x) dt \right] < +\infty, \quad \text{for each } x \in \mathbf{R}_+.$$

The following Theorem 5.1 is one of the main results in this section, which generalizes, to some extent, some corresponding existence results for bounded solutions of BSDEs with quadratic growth coefficients obtained respectively in Lepeltier and San Martin [16], Kobylanski [15], Briand and Hu [4, 5] and Fan [9]. We also point out that the usual PDE-based or ODE-based methods used in these references are not effective under the assumptions of Theorem 5.1.

Theorem 5.1. *Assume that $0 < T \leq +\infty$, $\xi \in L^\infty(\Omega, \mathcal{F}_T, P)$ and the generator g satisfies (H3) and (H4). Then BSDE (ξ, g) admits a bounded solution $(y_t, z_t)_{t \in [0, T]}$, and $\int_0^\cdot z_s \cdot dB_s$ is a BMO-martingale.*

To prove Theorem 5.1, we first establish the following two lemmas. We mention that Lemma 5.2 can be obtained by Theorems 3.1 and 4.1 in Fan [9] when $f, \bar{u} \in L^1([0, T]; \mathbf{R}_+)$.

Lemma 5.1. *Assume that $0 < T \leq +\infty$, $\xi \in L^\infty(\Omega, \mathcal{F}_T, P)$, the generator g satisfies (H3) with $f, \bar{u}$ and k , and (y, z) is a bounded solution of BSDE (ξ, g) . Then $\int_0^\cdot z_s \cdot dB_s$ is a BMO-martingale.*

Proof. Let Σ_T represent the set of all $(\mathcal{F}_t)$ -stopping time τ such that $0 \leq \tau \leq T$. Applying Itô's formula to $e^{3k|y \cdot|}$ and then taking conditional expectation with respect to $\mathcal{F}_\tau$ yield that for each $\tau \in \Sigma_T$, dP -a.s.,

$$\begin{aligned} & \frac{1}{2} \mathbb{E} \left[\int_\tau^T 9k^2 e^{3k|y_s|} |z_s|^2 ds \middle| \mathcal{F}_\tau \right] \\ & \leq \mathbb{E}[e^{3k|\xi|} | \mathcal{F}_\tau] + \mathbb{E} \left[\int_\tau^T 3k e^{3k|y_s|} \text{sgn}(y_s) g(s, y_s, z_s) ds \middle| \mathcal{F}_\tau \right]. \end{aligned}$$

Since $|y \cdot| \leq A$ with $A \in \mathbf{R}_+$ and g satisfies (H3) with $f, \bar{u}$ and k , we can deduce that, for each $\tau \in \Sigma_T$,

$$\begin{aligned} & \frac{3}{2} k^2 \mathbb{E} \left[\int_\tau^T e^{3k|y_s|} |z_s|^2 ds \middle| \mathcal{F}_\tau \right] \\ & \leq e^{3kA} + 3k e^{3kA} \mathbb{E} \left[\int_\tau^T (f_s + A \bar{u}_s) ds \middle| \mathcal{F}_\tau \right] \\ & \leq e^{3kA} + 3k e^{3kA} \left\{ \left\| \int_0^T f_t dt \right\|_\infty + A \left\| \int_0^T \bar{u}_t dt \right\|_\infty \right\}, \quad dP\text{-a.s.}, \end{aligned}$$

from which it follows immediately that

$$\sup_{\tau \in \Sigma_T} \left\| \mathbb{E} \left[\int_\tau^T |z_s|^2 ds \middle| \mathcal{F}_\tau \right] \right\|_\infty < +\infty.$$

The proof is then complete. $\square$

Lemma 5.2. Assume that $0 < T \leq +\infty$, $\xi \in L^\infty(\Omega, \mathcal{F}_T, P)$, and the generator

$$g'(\omega, t, y, z) := f_t(\omega) + \bar{u}_t(\omega)|y| + k|z|^2, \quad \forall \omega, t, y, z.$$

where $f, \bar{u} \in L^\infty(\Omega; L^1([0, T]; \mathbf{R}_+))$ and $k \in \mathbf{R}_+$. Then BSDE $(\|\xi\|_\infty, g')$ admits a unique bounded solution $(y'_t, z'_t)_{t \in [0, T]}$.

Furthermore, assume that the generator g satisfies (H3) with $f, \bar{u}$ and k , and that $(y_t, z_t)_{t \in [0, T]}$ is any bounded solution of BSDE (ξ, g) . Then for each $t \in [0, T]$, $|y_t| \leq y'_t$, dP -a.s.

Proof. Let $\bar{\xi} := \|\xi\|_\infty e^{\int_0^T \bar{u}_t dt} + \int_0^T e^{\int_0^t \bar{u}_s ds} f_t dt$. Note that for each $t \in [0, T]$, dP -a.s., $|ke^{-\int_0^t \bar{u}_s ds}| \leq k$ and

$$\begin{aligned} 0 &\leq \|\xi\|_\infty e^{\int_0^T \bar{u}_t dt} + \int_0^T e^{\int_0^t \bar{u}_s ds} f_t dt \\ &\leq \|\xi\|_\infty e^{\|\int_0^T \bar{u}_t dt\|_\infty} + \left\| \int_0^T f_t dt \right\|_\infty \cdot e^{\|\int_0^T \bar{u}_t dt\|_\infty} < +\infty. \end{aligned}$$

By Lemma 2.3 with $u(\cdot) = 0$ and $\gamma = 2k$, the following BSDE

$$\bar{y}'_t = \bar{\xi} + \int_t^T ke^{-\int_0^s \bar{u}_r dr} |\bar{z}'_s|^2 ds - \int_t^T \bar{z}'_s \cdot dB_s, \quad t \in [0, T] \quad (5.2)$$

admits a minimal bounded solution $(\bar{y}', \bar{z}')$. Now, define

$$y'_t := e^{-\int_0^t \bar{u}_s ds} \left(\bar{y}'_t - \int_0^t e^{\int_0^s \bar{u}_r dr} f_s ds \right) \quad \text{and} \quad z'_t := e^{-\int_0^t \bar{u}_s ds} \bar{z}'_t.$$

By Itô's formula, it is not hard to verify that

$$y'_t = \|\xi\|_\infty + \int_t^T (f_s + \bar{u}_s y'_s + k|z'_s|^2) ds - \int_t^T z'_s \cdot dB_s, \quad t \in [0, T].$$

Note that by (5.2) for each $t \in [0, T]$, dP -a.s., $\bar{y}'_t \geq \mathbb{E}[\bar{\xi} | \mathcal{F}_t]$, and then $y'_t \geq 0$, dP -a.s. It follows that $(y'_t, z'_t)_{t \in [0, T]}$ is a bounded solution of BSDE $(\|\xi\|_\infty, g')$. In addition, the uniqueness of bounded solution for BSDE $(\|\xi\|_\infty, g')$ is a direct consequence of Lemma 5.1 and Theorem 4.1.

Furthermore, assume that the generator g satisfies (H3) with $f, \bar{u}$ and k , and that (y, z) is a bounded solution of BSDE (ξ, g) . It follows from (H3) that $dP \times dt$ -a.e.,

$$\forall y > 0 \quad \text{and} \quad z \in \mathbf{R}^d, \quad g(\omega, t, y, z) \leq g'(\omega, t, y, z), \quad (5.3)$$

from which, since dP -a.s., $y'_t \geq 0$ for each $t \in [0, T]$, we know that

$$\mathbf{1}_{y_t > y'_t} (g(t, y_t, z_t) - g'(t, y_t, z_t)) \leq 0, \quad dP \times dt\text{-a.e.} \quad (5.4)$$

In view of the facts that both g and g' satisfy (H3) and $(y_t, z_t)_{t \in [0, T]}$ and $(y'_t, z'_t)_{t \in [0, T]}$ are, respectively, a bounded solution of BSDE (ξ, g) and BSDE $(\|\xi\|_\infty, g')$, it follows from Lemma 5.1 that both $\int_0^\cdot z_s \cdot dB_s$ and $\int_0^\cdot z'_s \cdot dB_s$ are

BMO-martingales. Thus, since dP -a.s., $\xi \leq \|\xi\|_\infty$, g' satisfies (H1) and (H2) with $h(\cdot) := k$, $v := 0$, and (5.4) holds true, by Theorem 4.1 we deduce that for each $t \in [0, T]$,

$$y_t \leq y'_t, \quad dP\text{-a.s.} \quad (5.5)$$

On the other hand, it follows from (H3) that $dP \times dt$ -a.e.,

$$\forall y < 0 \quad \text{and} \quad z \in \mathbf{R}^d, \quad -g'(\omega, t, y, z) \leq g(\omega, t, y, z).$$

Thus, with this inequality on hand, in view of Lemma 5.1 and Theorem 4.1, a similar argument to that from (5.3) to (5.5) yields that for each $t \in [0, T]$, $-y'_t \leq y_t$, dP -a.s. The desired conclusion follows. $\square$

We are now at a position to prove our existence result.

Proof. Assume that $0 < T \leq +\infty$, $\xi \in L^\infty(\Omega, \mathcal{F}_T, P)$ and the generator g satisfies (H3) with $f, \bar{u}, k$ and (H4) with $\psi(\cdot)$ and $\bar{h}(\cdot)$. We first consider the case when there exists a constant $p \in \mathbf{R}_+$ such that $dP \times dt$ -a.e., $g(\omega, t, y, z) \geq -pe^{-t}$ for each $(y, z) \in \mathbf{R} \times \mathbf{R}^d$. For each $(\omega, t, y, z) \in \Omega \times [0, T] \times \mathbf{R} \times \mathbf{R}^d$ and each $n \geq 1$, define:

$$g_n(\omega, t, y, z) := g(\omega, t, y, z) \wedge (ne^{-t}).$$

It is clear that $g_n \uparrow g$ and that for each $n \geq 1$, g_n is a continuous function of (y, z) which satisfies that

$$dP \times dt\text{-a.e.}, \quad |g_n(\omega, t, y, z)| = |g(\omega, t, y, z) \wedge (ne^{-t})| \leq (p+n)e^{-t}. \quad (5.6)$$

It is not hard to verify that g_n converges locally uniformly to g , i.e. for any sequences $\{(y^n, z^n)\}_{n=1}^{+\infty}$ in $\mathbf{R} \times \mathbf{R}^d$,

$$\text{if } (y^n, z^n) \rightarrow (y, z), \quad \text{then } dP \times dt\text{-a.e.}, \quad g_n(t, y^n, z^n) \rightarrow g(t, y, z), \quad \text{as } n \rightarrow +\infty. \quad (5.7)$$

In view of (5.6) and the fact of $g_n \uparrow g$, it follows from Lemma 2.3 that BSDE (ξ, g_n) admits a minimal bounded solution (y^n, z^n) for each $n \geq 1$, and that y^n is nondecreasing.

Furthermore, it is not hard to check from (H3) that $dP \times dt$ -a.e.,

$$\forall n \geq 1, \quad y \in \mathbf{R} \quad \text{and} \quad z \in \mathbf{R}^d, \quad \text{sgn}(y)g_n(\omega, t, y, z) \leq g'(\omega, t, y, z),$$

where

$$g'(\omega, t, y, z) := f_t(\omega) + \bar{u}_t(\omega)|y| + k|z|^2, \quad \forall \omega, t, y, z.$$

By Lemma 5.2, we know that for each $t \in [0, T]$ and $n \geq 1$, $|y_t^n| \leq y'_t \leq M$, dP -a.s., where (y', z') is the unique bounded solution of BSDE $(\|\xi\|_\infty, g')$ and M is a nonnegative constant. Define $y := \lim_{n \rightarrow \infty} y^n$. Then for each $t \in [0, T]$ and $n \geq 1$,

$$-M \leq -y'_t \leq y_t^n \leq y_t^{n+1} \leq y_t \leq y'_t \leq M, \quad dP\text{-a.s.} \quad (5.8)$$

In the sequel, we will use a localization procedure together with Proposition 3.1 to construct the desired solution. For each $l \geq 1$, define the following stopping time:

$$\tau_l(\omega) := \inf \left\{ t \in [0, T] : \int_0^t \psi_s(\omega, M) ds \geq l \right\} \wedge T.$$

Then $(y_l^n(t), z_l^n(t)) := (y_{t \wedge \tau_l}^n, z_t^n \mathbf{1}_{t \leq \tau_l})$ solves the following BSDE:

$$y_l^n(t) = y_{\tau_l}^n + \int_t^{\tau_l} \mathbf{1}_{s \leq \tau_l} g_n(s, y_l^n(s), z_l^n(s)) ds - \int_t^{\tau_l} z_l^n(s) \cdot dB_s. \quad (5.9)$$

In view of (5.8) and the facts that $|g_n| \leq g$ and g satisfies (H4) with $\psi(\cdot)$ and $\bar{h}(\cdot)$, we get that $dP \times ds$ -a.e., for each $(y, z) \in [-M, M] \times \mathbf{R}^d$,

$$\sup_{n \geq 1} \mathbf{1}_{s \leq \tau_l} |g_n(\omega, s, y, z)| \leq \mathbf{1}_{s \leq \tau_l} \psi_s(\omega, M) + \left(\max_{y \in [-M, M]} \bar{h}(y) \right) |z|^2. \quad (5.10)$$

Thus, in view of (5.8), (5.10), (5.7) and the definition of τ_l , by Proposition 3.1 we can take limit with respect to n (l being fixed) in (5.9) in the space $\mathcal{S}^\infty \times M^2$. In particular, for each $l \geq 1$, setting $y_l(t) := \lim_{n \rightarrow \infty} y_l^n(t)$, we know that $y_l(\cdot)$ is continuous and that there exists a process $z_l(t) \in M^2$ such that $\lim_{n \rightarrow \infty} z_l^n(t) = z_l(t)$ in M^2 and $(y_l(t), z_l(t))_{t \in [0, T]}$ solves the following BSDE:

$$y_l(t) = y_{\tau_l} + \int_t^{\tau_l} \mathbf{1}_{s \leq \tau_l} g(s, y_l(s), z_l(s)) ds - \int_t^{\tau_l} z_l(s) \cdot dB_s. \quad (5.11)$$

Since $\tau_l \leq \tau_{l+1}$, it follows from the definitions of $y_l(\cdot)$, $z_l(\cdot)$ and y , that

$$y_{t \wedge \tau_l} = y_{l+1}(t \wedge \tau_l) = y_l(t) = \lim_{n \rightarrow +\infty} y_{t \wedge \tau_l}^n, \quad z_{l+1}(t) \mathbf{1}_{t \leq \tau_l} = z_l(t) = \lim_{n \rightarrow +\infty} z_t^n \mathbf{1}_{t \leq \tau_l}.$$

Thus, since $y_l(\cdot)$ are continuous processes and moreover dP -a.s., $\tau_l = T$ for l large enough, it follows that y is continuous on $[0, T]$. Then we define z on $(0, T)$ by setting

$$z_t = z_l(t), \quad \text{if } t \in (0, \tau_l),$$

so that $z_t \mathbf{1}_{t \leq \tau_l} = z_l(t) \mathbf{1}_{t \leq \tau_l} = z_l(t)$ and (5.11) can be written as

$$y_{t \wedge \tau_l} = y_{\tau_l} + \int_{t \wedge \tau_l}^{\tau_l} g(s, y_s, z_s) ds - \int_{t \wedge \tau_l}^{\tau_l} z_s \cdot dB_s. \quad (5.12)$$

Furthermore, we have

$$\begin{aligned} & P \left(\int_0^T |z_s|^2 ds = +\infty \right) \\ &= P \left(\int_0^T |z_s|^2 ds = +\infty, \tau_l = T \right) + P \left(\int_0^T |z_s|^2 ds = +\infty, \tau_l < T \right) \\ &\leq P \left(\int_0^{\tau_l} |z_l(s)|^2 ds = +\infty \right) + P(\tau_l < T), \end{aligned}$$

and we deduce, in view of $\tau_l \uparrow T$, that

$$\int_0^T |z_s|^2 ds < +\infty, \quad dP\text{-a.s.}$$

Thus, letting $l \rightarrow +\infty$ in (5.12) yields that (y, z) is a solution of BSDE (ξ, g) . And, by (5.8) we know that $y \in \mathcal{S}^\infty$. On the other hand, in view of (5.8) and the fact that g satisfies (H3) with $f, \bar{u}$ and k , applying Itô's formula to $e^{3k|y_t|}$ on $[0, \sigma_l]$ where for each $l \geq 1$,

$$\sigma_l := \inf \left\{ t \in [0, T] : \int_0^t |z_s|^2 ds \geq l \right\} \wedge T,$$

by a similar argument to that in Lemma 5.1 and by virtue of Fatou's formula, we can deduce that

$$\begin{aligned} & \frac{3}{2}k^2 \mathbb{E} \left[\lim_{l \rightarrow +\infty} \int_0^{\sigma_l} e^{3k|y_s|} |z_s|^2 ds \right] \\ & \leq \lim_{l \rightarrow +\infty} \mathbb{E} [e^{3k|y_{\sigma_l}|}] + 3k \lim_{l \rightarrow +\infty} \mathbb{E} \left[\left(\int_0^{\sigma_l} e^{3k|y_s|} (f_s + \bar{u}_s |y_s|) ds \right) \right] \\ & \leq e^{3k|M|} + 3ke^{3k|M|} \left(\left\| \int_0^T f_t dt \right\|_\infty + |M| \left\| \int_0^T \bar{u}_t dt \right\|_\infty \right), \end{aligned}$$

from which it follows that $z \in M^2$. Then $(y_t, z_t)_{t \in [0, T]}$ is a bounded solution of BSDE (ξ, g) . Furthermore, it follows from Lemma 5.1 that $\int_0^\cdot z_s \cdot dB_s$ is a BMO-martingale.

Finally, for the general case, we approximate g by the sequence $\{g^{n,p}\}_{n,p}$, which are defined as follows:

$$g^{n,p}(\omega, t, y, z) := g^+(\omega, t, y, z) \wedge (ne^{-t}) - g^-(\omega, t, y, z) \wedge (pe^{-t}).$$

It follows from Lemma 2.3 that for each n and p there exists a minimal bounded solution $(y^{n,p}, z^{n,p}) \in \mathcal{S}^\infty \times M^2$ of BSDE $(\xi, g^{n,p})$. A similar argument as above, first letting $n \rightarrow +\infty$ and then $p \rightarrow +\infty$, shows that as $n, p \rightarrow +\infty$, $(y^{n,p}, z^{n,p})$ converges in $\mathcal{S}^\infty \times M^2$ to a bounded solution (y, z) of BSDE (ξ, g) , and $\int_0^\cdot z_s \cdot dB_s$ is a BMO-martingale. The proof of Theorem 5.1 is then complete. $\square$

Example 5.1. For $\xi \in L^\infty(\Omega, \mathcal{F}_T, P)$, we consider BSDE (ξ, g) with

$$g(\omega, t, y, z) := \mathbf{1}_{\Omega \times [0, 1]} e^{B_t^A(\omega)y^-} + \sqrt{u_t(\omega)|y|} \cdot |z| - v_t(\omega)e^y|z| + e^{-y}|z|^2,$$

where $u \in L^\infty(\Omega; L^1([0, T]; \mathbf{R}_+))$ and $v \in L^\infty(\Omega; L^2([0, T]; \mathbf{R}_+))$. It is not difficult to verify that g satisfies (H3) and (H4). It then follows from Theorem 5.1 that BSDE (ξ, g) admits a bounded solution. We mention that this conclusion cannot be obtained by any existing results.

The following existence and uniqueness result of the bounded solution for BSDEs is a direct consequence of Theorem 5.1, Lemma 5.1, Theorem 4.1 and Remark 5.1.

Theorem 5.2. Assume that $0 < T \leq +\infty$, $\xi \in L^\infty(\Omega, \mathcal{F}_T, P)$ and the generator g satisfies (H1), (H2) and (H4) with $g(\cdot, 0, 0) \in L^\infty(\Omega; L^1([0, T]; \mathbf{R}_+))$. Then BSDE (ξ, g) admits a unique bounded solution $(y_t, z_t)_{t \in [0, T]}$, and $\int_0^\cdot z_s \cdot dB_s$ is a BMO-martingale.

By Theorem 5.2, the following existence and uniqueness result of bounded solutions is immediate, which will be used in the next section.

Corollary 5.1. Assume that $0 < T \leq +\infty$, $b \in \mathbf{R}_+$, $f, u, \in L^\infty(\Omega; L^1([0, T]; \mathbf{R}_+))$ and $v, \in L^\infty(\Omega; L^2([0, T]; \mathbf{R}_+))$. Then the following BSDE

$$Y_t = b + \int_t^T (f_s + u_s |Y_s| + v_s |Z_s|) ds - \int_t^T Z_s \cdot dB_s, \quad t \in [0, T] \quad (5.13)$$

admits a unique bounded solution $(Y_t, Z_t)_{t \in [0, T]}$, which is also the unique L^2 solution (recall Lemma 2.1). Moreover, $(-Y_t, -Z_t)_{t \in [0, T]}$ is the unique bounded solution and L^2 solution of the following BSDE

$$\bar{Y}_t = -b - \int_t^T (f_s + u_s |\bar{Y}_s| + v_s |\bar{Z}_s|) ds - \int_t^T \bar{Z}_s \cdot dB_s, \quad t \in [0, T]. \quad (5.14)$$

Example 5.2. For $\xi \in L^\infty(\Omega, \mathcal{F}_T, P)$, $u, \in L^\infty(\Omega; L^1([0, T]; \mathbf{R}_+))$ and $v, \in L^\infty(\Omega; L^1([0, T]; \mathbf{R}_+)) \cap L^\infty(\Omega; L^2([0, T]; \mathbf{R}_+))$, we consider BSDE (ξ, g) with

$$g(\omega, t, y, z) := u_t(\omega)|y| - \mathbf{1}_{\Omega \times [0, 1]} e^{B_t^{(4)}(\omega)} y^5 + v_t(\omega) \sqrt{|z|} + \sqrt{v_t(\omega)} e^{-y} |z|^{\frac{3}{2}} - e^y |z|^2.$$

In view of Remark 4.2, it is easy to verify that this generator g satisfies (H1), (H2) and (H4) with $g(\cdot, 0, 0) \equiv 0$. It follows from Theorem 5.2 that BSDE (ξ, g) admits a unique bounded solution. Note that this g satisfies neither the conditions of Theorem 3.1 in Fan [9] nor the conditions of Theorem 2 in Lepeltier and San Martin [16]. Hence, the above conclusion cannot be obtained by these results.

Remark 5.3. We would like to mention that it is by far unknown that under the conditions of Theorem 5.1, whether BSDE (ξ, g) admits a minimal or maximal bounded solution. In the next section, we will introduce a stronger assumption (H5) (resp. (H6)) than (H3) and (H4), and establish a general existence result of the minimal (resp. maximal) bounded solution for BSDE (ξ, g) .

6. Existence of the Minimal (Maximal) Bounded Solution

In this section, by developing a method employed in Fan [9] and Lepeltier and San Martin [16] we will first prove a general existence theorem of the minimal (resp. maximal) bounded solution for BSDEs (see Theorem 6.1). Then, by virtue of Theorem 6.1, we will also give two comparison theorems of the minimal and maximal bounded solutions of BSDEs (see Theorems 6.2 and 6.3). Let us first introduce the following two assumptions, where $0 < T \leq +\infty$.

(H5) There exist three stochastic processes $\tilde{f}, \tilde{u}, \in L^\infty(\Omega; L^1([0, T]; \mathbf{R}_+))$, $\tilde{v}, \in L^\infty(\Omega; L^2([0, T]; \mathbf{R}_+))$ and a constant $k \in \mathbf{R}_+$ such that $dP \times dt$ -a.e.,

for each $(y, z) \in \mathbf{R}^{1+d}$,

$$-\tilde{f}_t(\omega) - \tilde{u}_t(\omega)|y| - \tilde{v}_t(\omega)|z| \leq g(\omega, t, y, z) \leq \tilde{f}_t(\omega) + \tilde{u}_t(\omega)|y| + \tilde{k}|z|^2.$$

(H6) There exist three stochastic processes $\tilde{f}, \tilde{u}, \tilde{v} \in L^\infty(\Omega; L^1([0, T]; \mathbf{R}_+))$, $\tilde{v} \in L^\infty(\Omega; L^2([0, T]; \mathbf{R}_+))$ and a constant $\tilde{k} \in \mathbf{R}_+$ such that $dP \times dt$ -a.e., for each $(y, z) \in \mathbf{R}^{1+d}$,

$$-\tilde{f}_t(\omega) - \tilde{u}_t(\omega)|y| - \tilde{k}|z|^2 \leq g(\omega, t, y, z) \leq \tilde{f}_t(\omega) + \tilde{u}_t(\omega)|y| + \tilde{v}_t(\omega)|z|.$$

The following Theorem 6.1 is the first result of this section, which generalizes partly Theorem 3.1 in Fan [9] and Theorem 2 in Lepeltier and San Martin [16].

Theorem 6.1. *Assume that $0 < T \leq +\infty$, $\xi \in L^\infty(\Omega, \mathcal{F}_T, P)$, and the generator g satisfies (H5) (resp. (H6)). Then BSDE (ξ, g) admits a minimal (resp. maximal) bounded solution $(y_t, z_t)_{t \in [0, T]}$, and $\int_0^\cdot z_s \cdot dB_s$ is a BMO-martingale.*

Proof. We only prove the case of the minimal solution, another case can be proved in a similar way. First, choose a sequence of smooth functions $\kappa_n(\cdot) : \mathbf{R}^d \mapsto [0, 1]$ such that

$$(1) \quad 0 \leq \kappa_n \leq 1; \quad (2) \quad \kappa_n(z) = 1 \text{ if } |z| \leq n; \quad (3) \quad \kappa_n(z) = 0 \text{ if } |z| \geq n + 1.$$

For each $(\omega, t, y, z) \in \Omega \times [0, T] \times \mathbf{R} \times \mathbf{R}^d$ and each $n \geq 1$, define:

$$g_n(\omega, t, y, z) := (-\tilde{f}_t(\omega) - \tilde{u}_t(\omega)|y| - \tilde{v}_t(\omega)|z|)(1 - \kappa_n(ze^{\frac{1}{2}})) + \kappa_n(ze^{\frac{1}{2}})g(\omega, t, y, z).$$

It is immediately seen from (H5) that $g_n \uparrow g$, and that g_n is a continuous function of (y, z) which satisfies that for each $n \geq 1$, $dP \times dt$ -a.e., $\forall y \in \mathbf{R}$ and $z \in \mathbf{R}^d$,

$$|g_n(\omega, t, y, z)| \leq \tilde{f}_t(\omega) + \tilde{u}_t(\omega)|y| + \tilde{v}_t(\omega)|z| + \tilde{k}(n+1)^2 e^{-t}. \quad (6.1)$$

It is not hard to verify that g_n converges locally uniformly to g , i.e. for any sequences $\{(y^n, z^n)\}_{n=1}^{+\infty}$ in $\mathbf{R} \times \mathbf{R}^d$,

$$\text{if } (y^n, z^n) \rightarrow (y, z), \quad \text{then } dP \times dt\text{-a.e., } g_n(t, y^n, z^n) \rightarrow g(t, y, z), \quad \text{as } n \rightarrow +\infty. \quad (6.2)$$

Obviously, in view of (6.1), it follows from Lemma 2.2 that the following BSDE

$$\theta_t^n = \xi + \int_t^T g_n(s, \theta_s^n, \Gamma_s^n) ds - \int_t^T \Gamma_s^n \cdot dB_s, \quad t \in [0, T]$$

admits a minimal L^2 solution (θ^n, Γ^n) . Furthermore, for each $n \geq 1$, $dP \times dt$ -a.e., $\forall y \in \mathbf{R}$ and $z \in \mathbf{R}^d$, we have

$$\begin{aligned} \lambda(\omega, t, y, z) &:= -\tilde{f}_t(\omega) - \tilde{u}_t(\omega)|y| - \tilde{v}_t(\omega)|z| \leq g_n(\omega, t, y, z) \\ &\leq \lambda_n(\omega, t, y, z) := \tilde{f}_t(\omega) + \tilde{u}_t(\omega)|y| + \tilde{k}|z|^2 \kappa_n(ze^{\frac{1}{2}}) \\ &\leq \lambda'(\omega, t, y, z) := \tilde{f}_t(\omega) + \tilde{u}_t(\omega)|y| + \tilde{k}|z|^2. \end{aligned} \quad (6.3)$$

By Corollary 5.1, BSDE $(-\|\xi\|_\infty, \lambda)$ admits a unique bounded solution (S, Z) , which is also the unique L^2 solution. It is not hard to verify that λ_n satisfies (A1)

and (A2) for each $n \geq 1$, and then, in view of Lemma 2.1, BSDE $(\|\xi\|_\infty, \lambda_n)$ admits a unique L^2 solution (Y^n, Z^n) . In addition, by virtue of Lemma 5.2, BSDE $(\|\xi\|_\infty, \lambda')$ admits a unique bounded solution (Q, Z') , which is also a L^2 solution. Thus, in view of (6.3) and the facts that both λ and λ_n satisfy (A1) for each $n \geq 1$, and $-\|\xi\|_\infty \leq \xi \leq \|\xi\|_\infty$, dP -a.s., it follows from Lemma 2.1 that for each $n \geq 1$ and $t \in [0, T]$,

$$S_t \leq \theta_t^n \leq Y_t^n \leq Q_t, \quad dP\text{-a.s.} \quad (6.4)$$

Note that both (S, Z) and (Q, Z') are bounded solutions and (6.4) holds true with $\sup_{t \in [0, T]}(|S_t| + |Q_t|) \leq A$ for some $A \in \mathbf{R}_+$. We know that for each $n \geq 1$, (θ^n, Γ^n) is a bounded solution of BSDE (ξ, g_n) with

$$\sup_{n \geq 1} |\theta_t^n| \leq A, \quad dP \times dt\text{-a.e.} \quad (6.5)$$

On the other hand, taking (6.1) and the fact of $g_n \leq g_{n+1}$ into account, we can conclude that θ^n is increasing by Lemma 2.2. Then we have the existence of θ such that for each $t \in [0, T]$,

$$\theta_t^n \uparrow \theta_t, \quad dP\text{-a.s.} \quad (6.6)$$

Moreover, by (6.3) we know that $dP \times dt$ -a.e.,

$$\forall (y, z) \in [-A, A] \times \mathbf{R}^d, \quad \sup_{n \geq 1} |g_n(\omega, t, y, z)| \leq f_t(\omega) + \mu|z|^2, \quad (6.7)$$

where $f_t(\omega) := \tilde{f}_t(\omega) + A\tilde{u}_t(\omega) + \tilde{v}_t^2(\omega) \in L^\infty(\Omega; L^1([0, T]; \mathbf{R}_+))$ and $\mu := 1 + \tilde{k}$. Thus, combining (6.5)–(6.7), by Proposition 3.1 we know that $\{(\Gamma_t^n)_{t \in [0, T]}\}_{n=1}^{+\infty}$ admits a convergence point Γ in M^2 , and in view of (6.2), that (θ, Γ) is a bounded solution of BSDE (ξ, g) . Furthermore, note that (H5) $\Rightarrow$ (H3). It follows from Lemma 5.1 that $\int_0^\cdot \Gamma_s \cdot dB_s$ is a BMO-martingale.

Finally, we show that (θ, Γ) is also the minimal bounded solution of BSDE (ξ, g) . Indeed, for any bounded solution $(\hat{\theta}, \hat{\Gamma})$ of BSDE (ξ, g) , which is also a L^2 solution, in view of (6.1) and the facts that $g_n \uparrow g$ and (θ^n, Γ^n) is the minimal L^2 solution of BSDE (ξ, g_n) , by Lemma 2.2 we can conclude that for each $n \geq 1$ and $t \in [0, T]$, $\theta_t^n \leq \hat{\theta}_t$, dP -a.s., and then $\theta_t \leq \hat{\theta}_t$ by (6.6). Theorem 6.1 is then proved. $\square$

The following Example 6.1 demonstrates that Theorem 6.1 generalizes some existing results.

Example 6.1. For $\xi \in L^\infty(\Omega, \mathcal{F}_T, P)$, $\tilde{u} \in L^\infty(\Omega; L^1([0, T]; \mathbf{R}_+))$ and $\tilde{v} \in L^\infty(\Omega; L^2([0, T]; \mathbf{R}_+))$, we consider BSDE (ξ, g) with

$$g(\omega, t, y, z) := \tilde{u}_t(\omega)y^{\frac{1}{3}} - \tilde{v}_t(\omega)|z| + e^{-|y|}|z|^2.$$

It is not difficult to verify that g satisfies (H5). It then follows from Theorem 6.1 that BSDE (ξ, g) admits a minimal bounded solution. Note that this g satisfies neither the conditions of Theorem 3.1 in Fan [9] nor the conditions of Theorem 2 in

Lepeltier and San Martin [16]. Hence, the above conclusion cannot be obtained by these results. In addition, we also mention that the bounded solution of this BSDE (ξ, g) is not necessarily unique.

Finally, by checking carefully the proof of Theorem 6.1 and making use of Lemma 2.2, we can get the following two comparison theorems for the minimal and maximal bounded solutions.

Theorem 6.2. *Assume that $0 < T \leq +\infty$, $\xi, \xi' \in L^\infty(\Omega, \mathcal{F}_T, P)$, both g and g' are two generators of BSDEs, and (y', z') is any bounded solution of BSDE (ξ', g') . Assume further that g satisfies (H5), and (y, z) is the minimal bounded solution of BSDE (ξ, g) (recall Theorem 6.1). If dP -a.s., $\xi \leq \xi'$, and $dP \times dt$ -a.e., $g(t, y, z) \leq g'(t, y, z)$ for each $(y, z) \in \mathbf{R}^{1+d}$, then for each $t \in [0, T]$, $y_t \leq y'_t$, dP -a.s.*

Theorem 6.3. *Assume that $0 < T \leq +\infty$, $\xi, \xi' \in L^\infty(\Omega, \mathcal{F}_T, P)$, both g and g' are two generators of BSDEs, and (y, z) is any bounded solution of BSDE (ξ, g) . Assume further that g' satisfies (H6), and (y', z') is the maximal bounded solution of BSDE (ξ', g') (recall Theorem 6.1). If dP -a.s., $\xi \leq \xi'$, and $dP \times dt$ -a.e., $g(t, y, z) \leq g'(t, y, z)$ for each $(y, z) \in \mathbf{R}^{1+d}$, then for each $t \in [0, T]$, $y_t \leq y'_t$, dP -a.s.*

Corollary 6.1. *Assume that $0 < T \leq +\infty$, $\xi, \xi' \in L^\infty(\Omega, \mathcal{F}_T, P)$, both generators g and g' satisfy (H5) (resp. (H6)), and (y, z) and (y', z') are, respectively, the minimal (resp. maximal) bounded solution of BSDE (ξ, g) and BSDE (ξ', g') (recall Theorem 6.1). If dP -a.s., $\xi \leq \xi'$, and $dP \times dt$ -a.e., $g(t, y, z) \leq g'(t, y, z)$ for each $(y, z) \in \mathbf{R}^{1+d}$, then for each $t \in [0, T]$, $y_t \leq y'_t$, dP -a.s.*

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